

Part (a): One-at-a-time hyperparameter sweeps (d8)

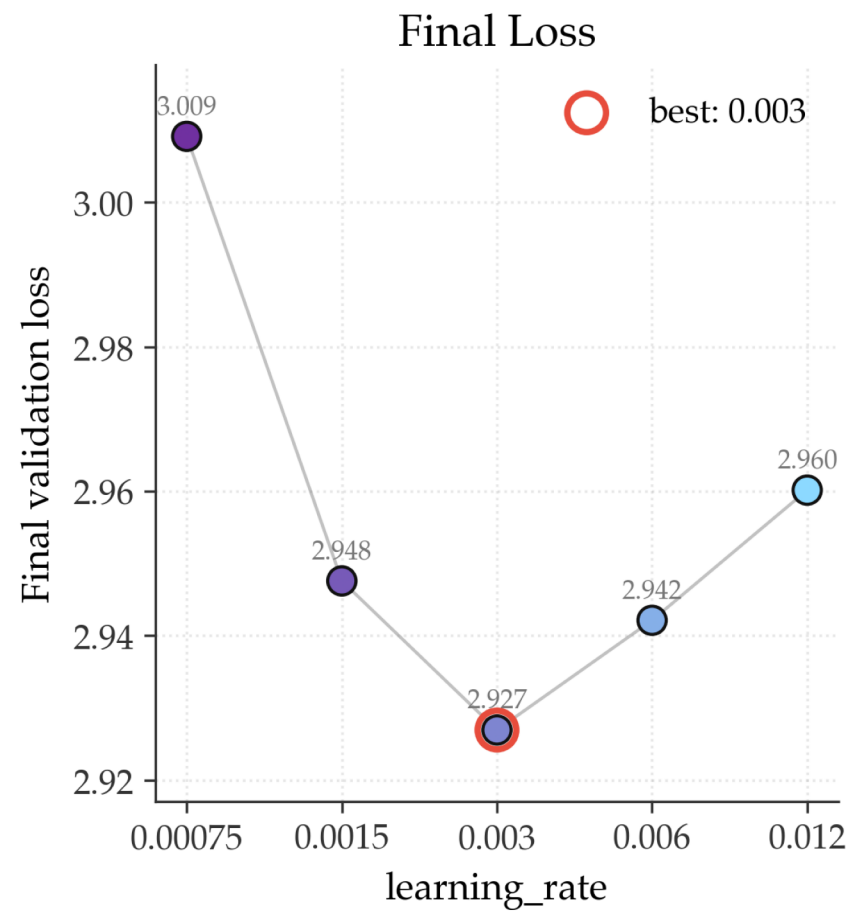
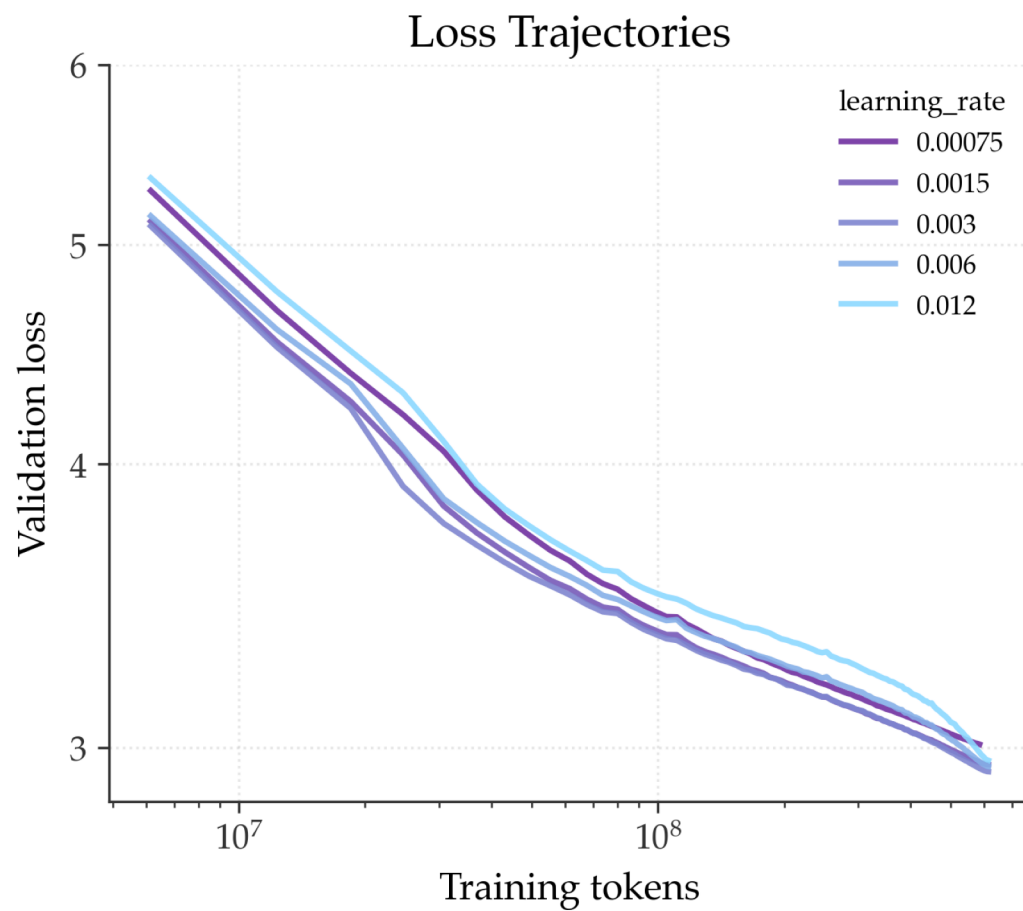
Compute-matched: d8, 614M tokens (600k seq x 1024). x2 log steps (/4 to x4) around defaults.
AdamW (0.9, 0.95), linear decay, grad clip 1.0, data seed 42, single model seed. Default run = center column.

	/4	/2	Default	x2	x4
Learning rate	7.5e-4	1.5e-3	3e-3	6e-3	1.2e-2
val loss	3.009	2.948	2.927	2.942	2.960
Batch size	16	32	64	128	256
val loss	2.928	2.919	2.927	2.950	3.013
Warmup %	0.0025	0.005	0.01	0.02	0.04
val loss	2.952	2.940	2.927	2.926	2.924
Weight decay	0.025	0.05	0.1	0.2	0.4
val loss	2.942	2.937	2.927	2.919	2.922

Importance (loss spread): batch size (0.094) ~ LR (0.082) >> warmup (0.028) ~ weight decay (0.023)

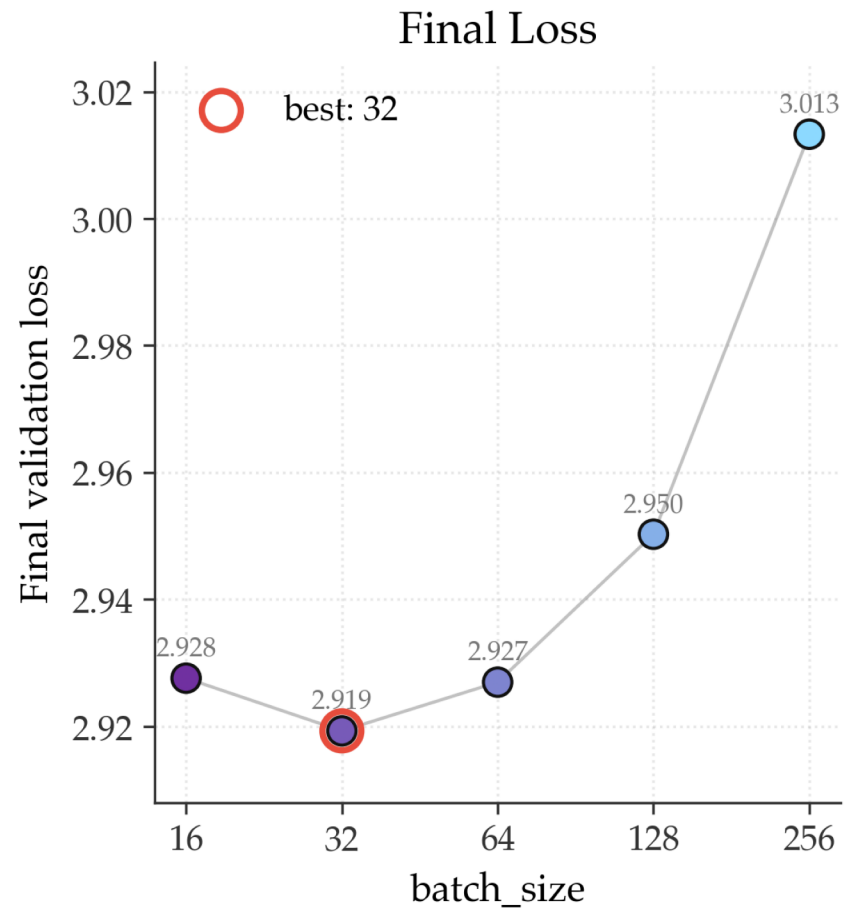
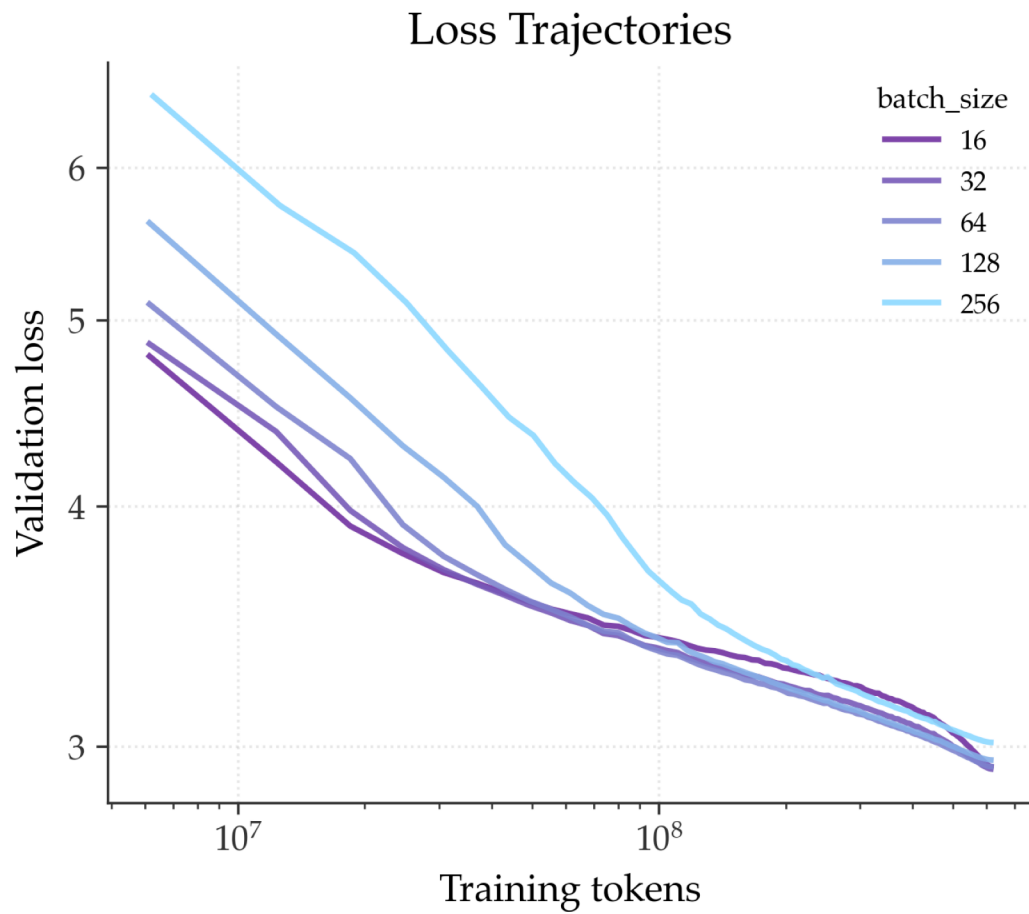
Source: W&B project [whitedeer-stanford-university/assignments](https://wandb.ai/whitedeer-stanford-university/assignments), tag hparam-sweeps-v2

Learning rate



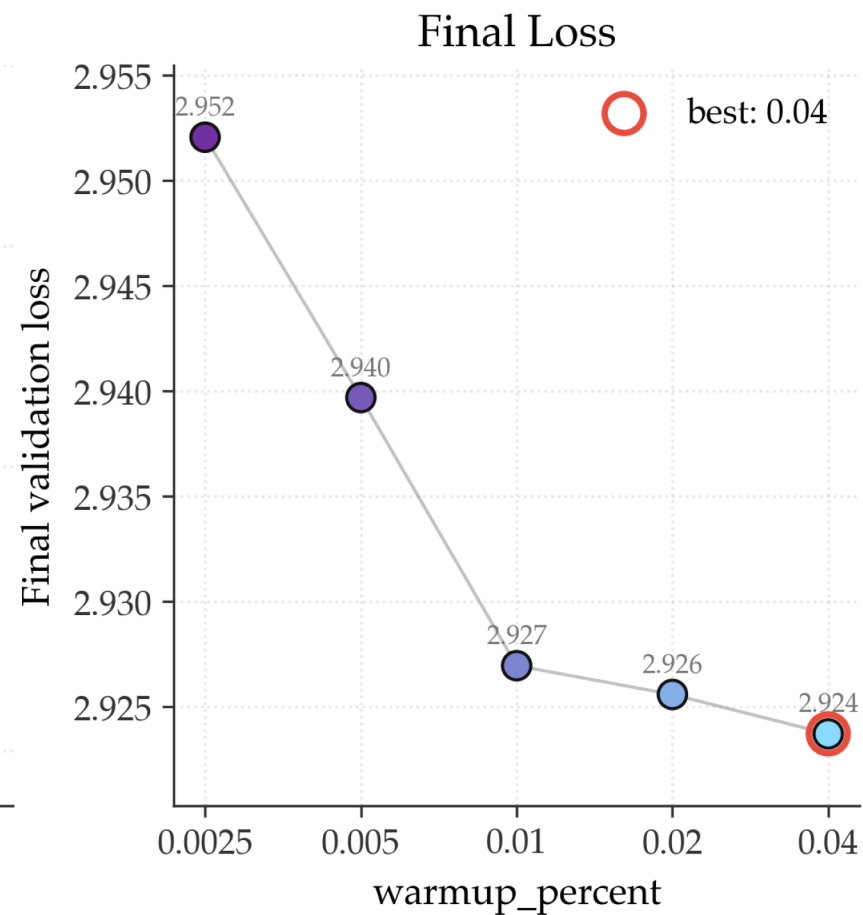
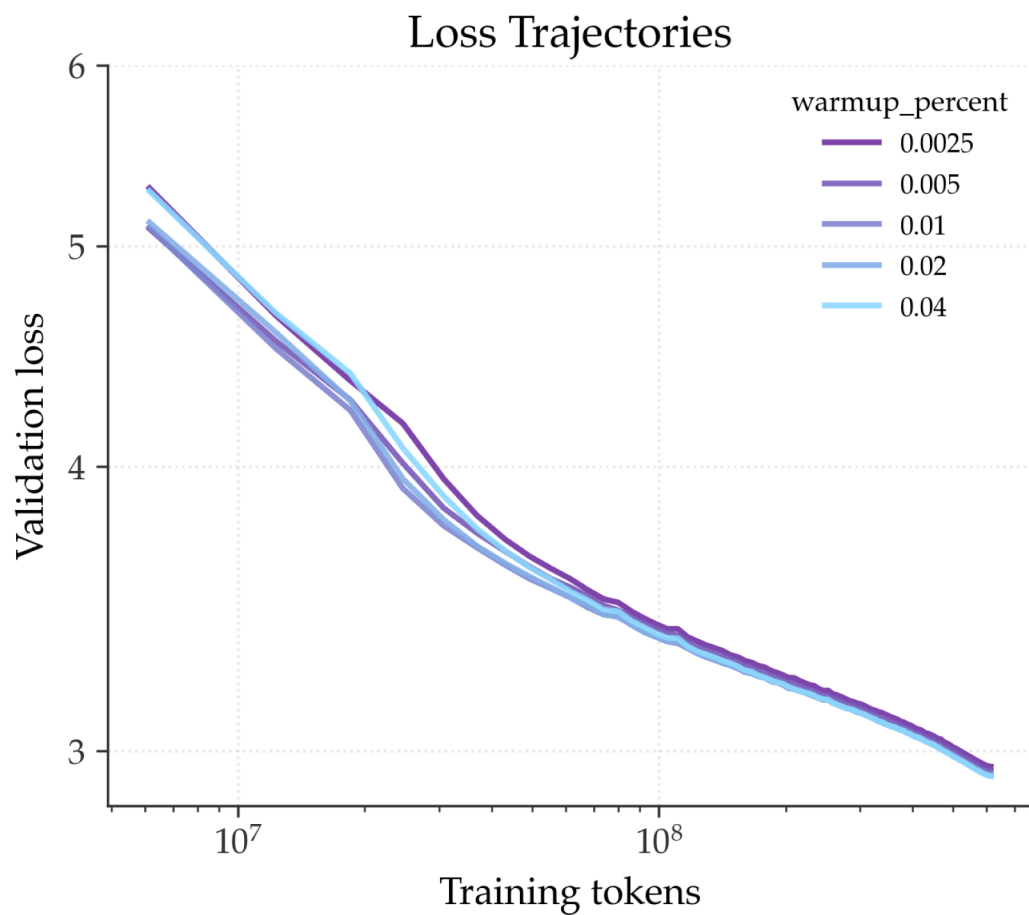
Best: 3e-3 (default). Spread 0.082. Too low hurts more than too high.

Batch size



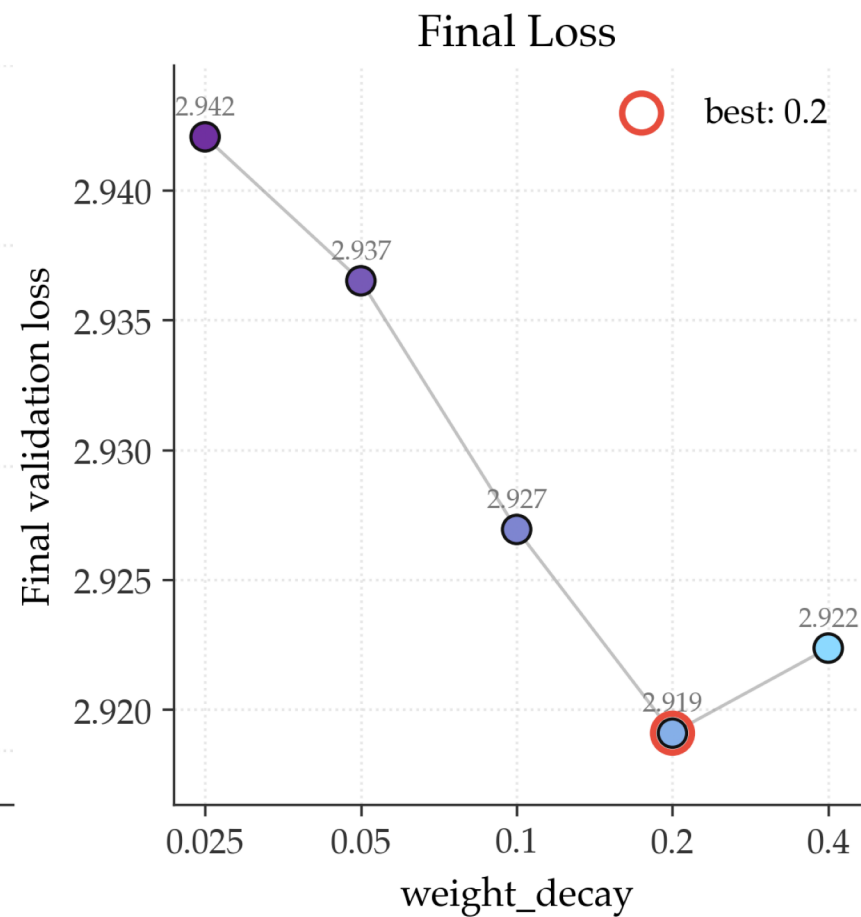
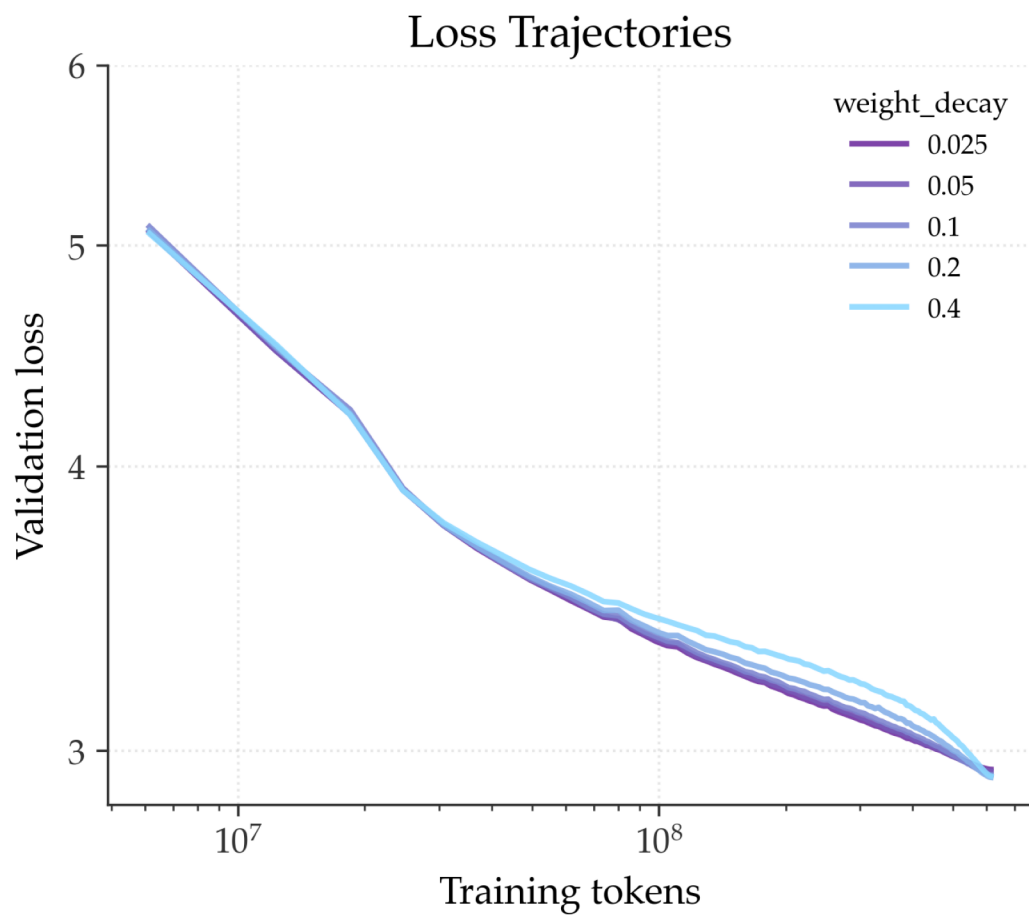
Best: 32. Spread 0.094. LR fixed at $3e-3$; large batches get too few steps.

Warmup %



Best: ≥ 0.04 (edge of range, minimum not bracketed). Spread 0.028.

Weight decay



Best: 0.2. Spread 0.023. Shallow minimum.